

Problem

Prove that the only solution in rational numbers of the equation

$$x^3 + 3y^3 + 9z^3 - 9xyz = 0$$

is

$$x = y = z = 0.$$

Source: 1983 Kürschák Competition

§1 A Descent Hidden Inside a Cyclic Transformation

We shall prove that the given equation admits no nonzero rational solution.

Assume, for contradiction, that

$$x^3 + 3y^3 + 9z^3 - 9xyz = 0$$

has a rational solution (x, y, z) with at least one nonzero coordinate.

Since the equation is homogeneous of degree three, multiplying by a suitable cube clears all denominators. Thus there exists a nonzero integer solution

$$(x_0, y_0, z_0) \in \mathbb{Z}^3$$

satisfying

$$x_0^3 + 3y_0^3 + 9z_0^3 - 9x_0y_0z_0 = 0.$$

Choose such a solution with

$$|x_0| + |y_0| + |z_0|$$

minimal among all nonzero integer solutions.

§2 The First Divisibility Step

Reducing the equation modulo 3, we obtain

$$x_0^3 \equiv 0 \pmod{3}.$$

Hence

$$3 \mid x_0.$$

Write

$$x_0 = 3x_1.$$

Substituting into the equation gives

$$27x_1^3 + 3y_0^3 + 9z_0^3 - 27x_1y_0z_0 = 0.$$

Dividing by 3 yields

$$y_0^3 + 3z_0^3 + 9x_1^3 - 9x_1y_0z_0 = 0.$$

Remarkably, this has exactly the same form as the original equation, with the variables cyclically shifted. Therefore

$$(y_0, z_0, x_1)$$

is another integer solution.

§3 Repeating the Argument

Applying the same reasoning to the new solution, reduction modulo 3 shows that

$$3 \mid y_0.$$

Hence

$$y_0 = 3y_1.$$

Substituting into the transformed equation and dividing by 3 once again, we obtain another solution

$$(z_0, x_1, y_1).$$

Repeating the argument a third time gives

$$3 \mid z_0,$$

so that

$$z_0 = 3z_1.$$

Consequently,

$$(x_0, y_0, z_0) = (3x_1, 3y_1, 3z_1).$$

Thus every coordinate of our integer solution is divisible by 3.

§4 The Infinite Descent

Dividing the original equation by 27 shows that

$$(x_1, y_1, z_1)$$

is again an integer solution of the same equation.

Moreover,

$$|x_1| + |y_1| + |z_1| = \frac{|x_0| + |y_0| + |z_0|}{3},$$

which is strictly smaller than

$$|x_0| + |y_0| + |z_0|$$

unless all coordinates are zero.

This contradicts the minimality of the chosen nonzero solution.

Equivalently, the above argument may be repeated indefinitely, forcing the solution to be divisible by arbitrarily high powers of 3, which is impossible for a nonzero integer triple.

Therefore no nonzero integer solution exists.

§5 The Final Contradiction

Since every rational solution gives rise to an integer solution, and no nonzero integer solution exists, the original equation has only the trivial rational solution.

Hence

$$\boxed{x = y = z = 0}.$$